

Lodestar: a guided walkthrough

A supply-chain command centre prototype for medical imaging · about 12 minutes · MR, then CT, then ultrasound

Getting ready

A few minutes before you start, click each of these four links. Each opens in its own browser tab, already set to the right product line and map view. Leave them open and move through the tabs in order as you present.

1. All product lines

overview

2. MR

helium and Hormuz

3. CT

export controls

4. Ultrasound

a calm day

- In each tab, click the **Layers** header (bottom left of the map) to fold that panel away.
- The data refreshes through the day, so the numbers you see will differ from the examples in this guide. Read them off the screen.

Opening

about 45 seconds

"What you're about to see is Lodestar, a working prototype of a supply-chain command centre for medical imaging.

It answers one question every morning: what is happening in the world right now that could stop us building, shipping or installing a system, and what should we decide about it today?

Everything outside our walls is real public data: shipping transits, export controls, earthquakes, travel advisories, FDA notices. Each item is time-stamped and linked to its source. Everything inside our walls (stock cover, installs, revenue) is synthetic for now, and it's labelled that way on every screen.

The AI reads the evidence and recommends. People decide.

I'll show you two product lines where the picture is most interesting, MR and CT, and then one where it's calm."

1. All product lines

1 minute · tab 1

The map:

- The **glow** shows exposure. Only items scoring 35 or more glow, and the number on a marker is its score out of 100.
- The **lines** are shipping lanes. Sea lanes follow real routes through the chokepoints they cross; air lanes are arcs.
- **Ports** are labelled, for example Charleston, with lanes to the UK, the Gulf and Brazil.

Hover over the Strait of Hormuz marker. The tooltip shows the latest daily count of ships through the strait, from IMF PortWatch.

"In 2025 this strait averaged about 85 ships a day. Look at it now."

The Command brief (right-hand side):

- The large figure is revenue at risk over 90 days. It is **synthetic**, and labelled so.
- The brief itself is written by the AI from the evidence. The model name and cost are shown.
- Each decision card has an owning function, a decide-by date and options, with the recommended option first.

"The model recommends; people decide."

2. MR

4 minutes · tab 2

Product exposure board: the conventional magnet scores high (around 70); the sealed low-helium magnet scores low (around 8).

"The sealed magnet uses under 1% of the helium, so it barely registers. That's the hedge, and you can see it."

Hotspots worth diving into: the Strait of Hormuz and the Qatar and Russian helium sources are at the top.

Click "Strait of Hormuz", then "Open evidence drawer". Walk down it from the top:

1. **The AI's read:** a headline and three sentences written from the evidence underneath.
2. **Chatter:** recent Telegram posts from news and open-source channels. Labelled unverified, and never used in the score.
3. **Why this was flagged:** each piece of evidence, tagged as live, reference or synthetic, with a bar for how strongly it counts.
4. **How it reaches us:** from the chokepoint, to the lanes through it, to helium, to the plants, products and installs.
5. **Survive vs recover:** how many days our stock lasts against how long supply takes to recover (around 41 against 120, synthetic). Drag the **what-if** slider from 2 months to 3 and watch installs short and revenue at risk climb. The formula is printed underneath.
6. **What we can do:** costed options, recommended first, with a decide-by date and an owner.

Under "Have the agent draft it", click "S&OP escalation"

A draft note appears in a few seconds. Point at the banner: **"needs your approval · nothing has been sent"**. Nothing is ever sent from Lodestar; "Approve & route" is for show only.

3. CT

3 minutes · tab 3

Click "← Back to map" to close the drawer, then switch to the CT tab.

- **Product board:** CT is also exposed to Hormuz, through the Beijing → Gulf shipping lane. Photon-counting CT is driven by tungsten.
- **Input clock:** tungsten is **exposed** (stock runs out before supply recovers), while gadolinium is **covered**.

"Both are under China's export controls. Only one is a problem, because the stock cover differs."

- **Open the Gadolinium drawer** from the hotspots list. The control line notes that China's October 2025 expansion is suspended until about 10 November 2026.

"That's the date to watch."

- **Regulatory & trade watch:** live FDA recalls and clearances, and the export controls on critical inputs, each tagged sourced or estimate.
- On the map, the **Critical Minerals** layer shows Bayan Obo, the world's largest rare-earth mine, north of our gadolinium source at Baotou.

4. Ultrasound

30 seconds · tab 4

Ultrasound scores low, with nothing at risk. The brief calls it a calm day and makes no decisions.

"It also tells you what not to worry about."

Closing

"Everything outside our walls comes from public feeds, time-stamped and refreshed every few hours. Connect ERP, the install backlog and the supplier master, and the synthetic numbers become real."

Good to know when answering questions

- **Synthetic vs real:** stock cover, installs, revenue and plant figures are illustrative, and marked "synth" wherever they appear. Everything external is public data with its source and date.
- **Shipping data is daily**, not live ship tracking; the counts are published a couple of days behind.
- **The AI is advisory.** It can make mistakes, so every recommendation shows its evidence for someone to check before acting.
- **Telegram posts are unverified** and shown for context only.